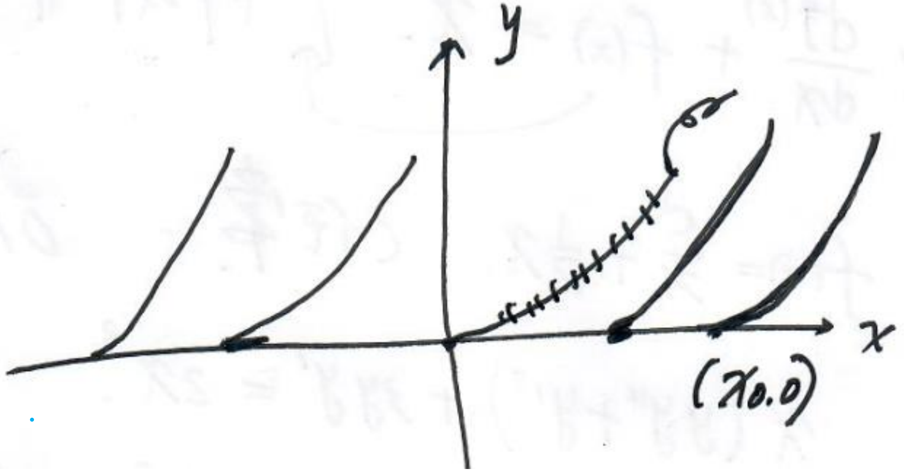
1. 求 $\frac{dy}{dx}=y^{\frac{1}{4}}$ 的通解，做积分曲线族的草图，

如有奇解，说明并解释.

解. $\frac{dy}{dx}=y^{1/4}$. $y(x)\equiv 0$为特解. $y\geq 0$. 定义域: $x\in(-\infty,+\infty)$,

且 $\frac{dy}{dx}=y^{1/4}>0$ (除 $y=0$).

由变量分离法得: $y^{3/4}=(x+c)\cdot\frac{3}{4}$, c任常.



任取 $x_0\in R$，原方程过 $(x_0,0)$ 的积分曲线为

$$y^{3/4}=\frac{3}{4}(x-x_0)\text{，且}\left.\frac{dy}{dx}\right|_{(x_0,0)}=\left.\left[\frac{3}{4}(x-x_0)\right]^3\right|_{x=x_0}=0,$$

$(x\geq x_0)$

由奇解的定义知，$y(x)\equiv 0$ 是奇解。 OK.

2. 已知 $(x^3+y)dx+f(x)dy=0$ 有积分因子 $\mu=x$. (P2)

$f(x)$连续可导, 求所有可能的$f(x)$.

解: $\because$ $x(x^3+y)dx+xf(x)dy=0$ 为恰当eq.

$$\therefore \frac{\partial(x^4+xy)}{\partial y}=\frac{\partial(xf(x))}{\partial x},$$

$$\Leftrightarrow x=f(x)+xf'(x).$$

即 $x\frac{df(x)}{dx}+f(x)=x$. [一阶线性非齐次]

$\Rightarrow f(x)=\frac{c}{x}+\frac{1}{2}x$, c任常. OK.

3. 求解 $x(yy''+y'^2)+3yy'=2x^3$.

解. 令 $z=yy'$, 则 $\frac{dz}{dx}=(y')^2+yy''$.

$\Rightarrow x\frac{dz}{dx}+3z=2x^3$ (二阶线性非齐次)

一阶

$\Rightarrow z(x)=\frac{c_1}{x^3}+\frac{1}{3}x^3$.

于是 $y\frac{dy}{dx}=\frac{c_1}{x^3}+\frac{1}{3}x^3$. 变量分离后通解为

$$y^2(x) = -\frac{c_1}{x^2} + \frac{1}{6}x^4 + c_2,\quad c_i\ (i=1,2)\ \text{任常.}\quad \text{OK.}$$

4. 求解方程组.

$$\frac{dy}{dx} = \begin{pmatrix} x^2 & 0 & 1 \\ 0 & \sin x & 1 \\ 0 & 0 & e^x \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} + \begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix},\quad y(0) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

解: 原方程等价于

$$\begin{cases} \dfrac{dy_1}{dx} = x^2 y_1 + y_3 + x, & ① \\[2ex] \dfrac{dy_2}{dx} = (\sin x)\, y_2 + y_3 & ② \\[2ex] \dfrac{dy_3}{dx} = e^x y_3 & ③ \end{cases}$$

由③ $\Rightarrow$ $$y_3(x) = c_3 e^{e^x}.\quad \text{由} y_3(0)=1 \text{ 得 } c_3 = e^{-1}$$

$$\therefore y_3(x) = e^{e^x - 1}.$$

由② $\Rightarrow$ $$\frac{dy_2}{dx} = \sin x \, y_2 + e^{e^x - 1}.$$

$$\Rightarrow y_2(x) = e^{-\cos x}\left(e + \frac{1}{e}\int_0^x \left(e^{e^s + \cos s}\right) ds\right).$$

由① 得.

$$y_1(x) = e^{\frac{1}{3}x^3}\left[1 + \int_0^x \left(e^{e^s - \frac{1}{3}s^3 - 1} + s e^{-\frac{1}{3}s^3}\right) ds\right].$$

P4 ok.

5. 求解 $\dfrac{d}{dx}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} \alpha & -2 \\ 2 & \alpha \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$, $\alpha \in R$.

并讨论当 $x \to +\infty$ 时解的发展趋势. 以及 2π 周期解的存在性. $(i^2 = -1)$

解: $|(\lambda E - A)| = (\lambda - \alpha)^2 + 4 = 0 \Rightarrow \lambda_{1,2} = \alpha \pm 2i$

$\alpha + 2i$ 对应的特征向量为 $\begin{pmatrix} i \\ 1 \end{pmatrix}$.

得 $e^{(\alpha+2i)x}\begin{pmatrix} i \\ 1 \end{pmatrix}$

$= e^{\alpha x}(\cos 2x + i\sin 2x)\begin{pmatrix} i \\ 1 \end{pmatrix}$

$= e^{\alpha x}\left[\begin{pmatrix} -\sin 2x \\ \cos 2x \end{pmatrix} + i\begin{pmatrix} \cos 2x \\ \sin 2x \end{pmatrix}\right]$

通解为

$\begin{pmatrix} y_1(x) \\ y_2(x) \end{pmatrix} = C_1 e^{\alpha x}\begin{pmatrix} -\sin 2x \\ \cos 2x \end{pmatrix} + C_2 e^{\alpha x}\begin{pmatrix} \cos 2x \\ \sin 2x \end{pmatrix}$, C_i 任常

(1) 当 $\alpha=0$：$\begin{cases} y_1(x) = -c_1\sin 2x + c_2\cos 2x \\ y_2(x) = c_1\cos 2x + c_2\sin 2x \end{cases}$

基解矩阵 $\Phi(x) = \begin{pmatrix} -\sin 2x & \cos 2x \\ \cos 2x & \sin 2x \end{pmatrix}$.

由于 $\Phi(\overset{x+}{\pi}) = \Phi(x)$，$\forall x \in R$. 所以 $\Phi(x)$ 是 π 周期的函数矩阵，~~从~~ 从而原eq的所有解均是 π 周期解.

② 当 $\alpha<0$ 时, $\lim\limits_{x\to+\infty}\begin{pmatrix}y_1(x)\\ y_2(x)\end{pmatrix}=0$, $\forall\, C_1, C_2$;

当 $\alpha>0$ 时, 由于 $\lim\limits_{k\to+\infty}\left\|e^{\alpha x_k}\begin{pmatrix}-\sin 2x_k\\ \cos 2x_k\end{pmatrix}\right\|$

$=+\infty$, 所以原eq存在无界解. 其中

$x_k=k\pi+\frac{\pi}{4}$, $k\in N$. OK.

(16)

8. 设 $y=\phi(x)$ 满足 $y'+a(x)y \le 0$ for $x \ge 0$.

$$\Rightarrow \phi(x) \le \phi(0)\exp\left(-\int_0^x a(s)ds\right) \text{ for } x \ge 0.$$

证明: For $x \ge 0$, 有

$$\phi'(x)+a(x)\phi(x) \le 0$$

$$\Rightarrow \underline{e^{\int_0^x a(s)ds}\phi'(x)+a(x)e^{\int_0^x a(s)ds}\phi(x)} \le 0$$

$$\|$$

$$\frac{d}{dx}\underbrace{e^{\int_0^x a(s)ds}\phi(x)}_{Z(x)} \le 0, \text{ for } x \ge 0 \quad (x \ge 0)$$

$\Rightarrow$ $Z(x)$ 在 $[0,+\infty)$ 上单减, 于是有 $Z(x) \le Z(0)$

即 $e^{\int_0^x a(s)ds}\phi(x) \le \phi(0)$. $(x \ge 0)$

$\Rightarrow$ ok.

1. $\begin{cases} \dfrac{dx}{dt} = y + x^2 \\ \dfrac{dy}{dt} = -x \end{cases}$ $\quad \begin{cases} x(0)=1 \\ y(0)=2. \end{cases}$

$$\begin{pmatrix} x_k(t) \\ y_k(t) \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} \int_0^t [y_{k-1}(s) + x_{k-1}^2(s)]\, ds \\ \int_0^t [-x_{k-1}(s)]\, ds \end{pmatrix}, \quad k=1,2,3,\cdots$$

$$n=0 \quad \begin{pmatrix} x_0(t) \\ y_0(t) \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

$$n=1. \quad \begin{pmatrix} x_1(t) \\ y_1(t) \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} f_1(t) \\ f_2(t) \end{pmatrix} = \begin{pmatrix} 1+f_1(t) \\ 2+f_2(t) \end{pmatrix}, = \begin{pmatrix} 1+3t \\ 2-t \end{pmatrix}.$$

$$f_1(t)=\int_0^t [2+1]\,ds = \underline{3t},$$

$$f_2(t)=\int_0^t [-1]\,ds = \underline{-t}.$$

$n=2.$

$$\begin{pmatrix} x_2(t) \\ y_2(t) \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} \overset{f_{21}(t)}{\cancel{g_1(t)}} \\ \underset{f_{22}(t)}{\cancel{g_2(t)}} \end{pmatrix} = \begin{pmatrix} \cancel{1+g_1(t)} \\ \cancel{2+g_2(t)} \end{pmatrix} = \begin{pmatrix} 1+f_{21}(t) \\ 2+f_{22}(t) \end{pmatrix}$$

$$\cancel{g_1(t)=\int_0^t} \quad f_{21}(t)=\int_0^t [y_1(s)+x_1^2(s)]\,ds$$

$$=\int_0^t [(2-s)+(1+3s)^2]\,ds = 3t+\frac{5}{2}t^2+3t^3$$

$$\Rightarrow x_2(t) = 1+3t+\frac{5}{2}t^2+3t^3.$$

$$f_{22}(t) = \int_0^t \left(-\underline{x_1(s)}\right) ds$$

$$= \int_0^t -(1+3s)\,ds = -t - \frac{3}{2}t^2$$

$$\Rightarrow y_2(t) = 2 - t - \frac{3}{2}t^2$$

$$\therefore \begin{pmatrix} x_2(t) \\ y_2(t) \end{pmatrix} = \begin{pmatrix} 1+3t+\frac{5}{2}t^2+3t^3 \\ 2-t-\frac{3}{2}t^2 \end{pmatrix}.$$

OK.

2. $\begin{cases} \dfrac{dx}{dt} = -y \\ \dfrac{dy}{dt} = x \end{cases} \qquad \begin{cases} x(0) = x_0 \\ y(0) = y_0 \end{cases}$

$$\begin{pmatrix} x_k(t) \\ y_k(t) \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} \int_0^t -y_{k-1}(s)\,ds \\ \int_0^t x_{k-1}(s)\,ds \end{pmatrix}, \qquad k = 1, 2, 3, \cdots$$

$n=0$: $\begin{pmatrix} x_0(t) \\ y_0(t) \end{pmatrix} = \boxed{\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}}$

$n=1$: $\begin{pmatrix} x_1(t) \\ y_1(t) \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} \int_0^t -y_0\,ds \\ \int_0^t x_0\,ds \end{pmatrix} = \begin{pmatrix} x_0 - \boxed{y_0 t} \\ y_0 + \boxed{t x_0} \end{pmatrix}$

$n=2$. $\begin{pmatrix} x_2(t) \\ y_2(t) \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} \int_0^t (-y_0 - s x_0)\,ds \\ \int_0^t (x_0 - y_0 s)\,ds \end{pmatrix} = \begin{pmatrix} x_0 - \frac{t^2}{2} x_0 - y_0 t \\ y_0 + x_0 t - \frac{t^2}{2} y_0 \end{pmatrix}$ P_2

$$= \begin{pmatrix} x_0 - y_0 t - \frac{t^2}{2} x_0 \\ y_0 + x_0 t - \frac{t^2}{2} y_0 \end{pmatrix} = x_0 \begin{pmatrix} 1 - \frac{t^2}{2} \\ t \end{pmatrix} + y_0 \begin{pmatrix} -t \\ 1 - \frac{t^2}{2} \end{pmatrix}$$

$n=3$.

$$\begin{pmatrix} x_3(t) \\ y_3(t) \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} \int_0^t -\left[x_0 s + y_0\left(1-\frac{s^2}{2}\right)\right] ds \\ \int_0^t \left[x_0\left(1-\frac{s^2}{2}\right) - y_0 s\right] ds \end{pmatrix}$$

$$= \begin{pmatrix} x_0\left[1 - \frac{t^2}{2}\right] + y_0\left[t - \frac{t^3}{3!}\right] \\ y_0\left[1 - \frac{t^2}{2}\right] + x_0\left[t - \frac{t^3}{3!}\right] \end{pmatrix}$$

$$= x_0 \begin{pmatrix} 1 - \frac{t^2}{2} \\ t - \frac{t^3}{3!} \end{pmatrix} + y_0 \begin{pmatrix} t - \frac{t^3}{3!} \\ 1 - \frac{t^2}{2} \end{pmatrix}$$

$n=4$:

$$\begin{pmatrix} x_4(t) \\ y_4(t) \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} \int_0^t -\left(y_0\left(1-\frac{s^2}{2}\right) - x_0\left(t-\frac{t^3}{3!}\right)\right) dt \\ \int_0^t \left[x_0\left(1-\frac{t^2}{2}\right) + y_0\left(t-\frac{t^3}{3!}\right)\right] dt \end{pmatrix}$$

$$= \begin{pmatrix} x_0 - x_0\left(\frac{t^2}{2} - \frac{1}{4!}t^4\right) - y_0\left(t-\frac{t^3}{3!}\right) \\ y_0\left[1+\frac{t^2}{2} - \frac{t^4}{4!}\right] + x_0\left[t - \frac{t^3}{3!}\right] \end{pmatrix}$$

$$= x_0 \boxed{\begin{pmatrix} 1-\frac{t^2}{2!}+\frac{1}{4!}x^4 \\ t-\frac{t^3}{3!} \end{pmatrix}} + y_0 \boxed{\begin{pmatrix} -\left(t-\frac{t^3}{3!}\right) \\ 1+\frac{t^2}{2}-\frac{t^4}{4!} \end{pmatrix}} \overset{\text{近似}}{=} x_0\begin{pmatrix} \cos t \\ \sin t \end{pmatrix} + y_0\begin{pmatrix} -\sin t \\ \cos t \end{pmatrix}.$$

OK.